Assessing and Forecasting SO_2 Pollution in the Continental United States : A Data-Driven Approach to Mitigating Health Risks

Jenna Lee
University of Hawai'i at Manoa

INTRODUCTION

Air quality deterioration caused by sulfur dioxide (SO_2) emissions remains a significant environmental and public health concern. SO_2, a primary air pollutant generated from industrial processes, fossil fuel combustion, and power generation, contributes to the formation of particulate matter ($PM_{2.5}$) and acid rain, further worsening air quality. Elevated SO_2 levels are known to trigger and exacerbate a range of respiratory diseases, including asthma, bronchitis, and other chronic lung conditions.

Vulnerable populations, such as children, the elderly, and individuals with pre-existing respiratory or cardiovascular conditions, are particularly at risk. The situation is further amplified in densely populated urban areas, where frequent and intense exposure to SO_2 and associated pollutants increases the likelihood of adverse health outcomes.



Moreover, SO_2 emissions contribute to climate forcing and environmental degradation, creating a feedback loop that affects both local ecosystems and global atmospheric conditions. Addressing this issue requires a comprehensive understanding of SO_2 concentration trends, population exposure patterns, and mitigation strategies.

This study aims to analyze SO_2 emissions, assess their impact on human health, and highlight the importance of environmental policies in reducing exposure, particularly in regions with high population densities.

OBJECTIVE

Research Questions:

- How do meteorological and environmental factors influence SO_2 concentration levels in the U.S?
- How accurately can we predict future SO_2 concentration levels in different region of U.S using historical data and meteorological conditions?
- What is the estimated exposure of the U.S population to harmful levels of SO_2 and which populations are most at risk?

METHOD

• Data Preprocessing

- Data Integration: Collected and combined raw SO_2 emission data from CMIP6 (emiso2 variable) and other atmospheric datasets.
- Cleaning & Standardization: Removed outliers and filled missing values to ensure consistency across sources. Data was standardized to account for differences in scales and units.

• Predictive Modeling

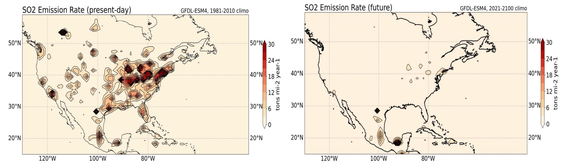
- Feature Selection: Identified key predictors, including historical SO_2 levels, and geographical data.
- Machine Learning Models: Developed and trained models such as Random Forest and Linear Regression to forecast SO_2 concentrations. Model accuracy was validated using cross-validation techniques.
- Scenario Analysis: Incorporated future emission scenarios (e.g., SSP585) to predict SO_2 trends under varying climate and policy conditions.

• Risk Analysis

- Population Exposure Assessment: Mapped SO_2 levels against demographic data to estimate exposure for different regions and population groups.
- Health Impact Evaluation: Quantified potential health risks (e.g., respiratory issues) with a focus on vulnerable populations such as children, the elderly, and those in low-income communities.
- Policy Implications: Analyzed results to provide recommendations for emission controls and public health interventions.

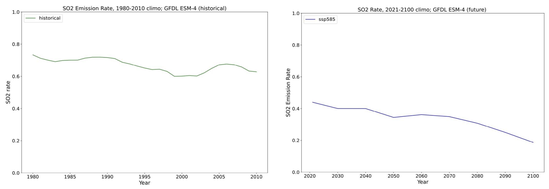
RESULTS

- Annual Mean of SO_2 (Future vs Present)



We can see that SO_2 emissions in the eastern United States have decreased significantly. You can see a decrease of more than 20 tons per square mile per year.

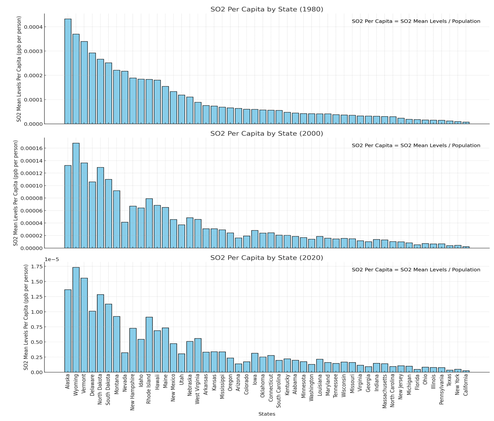
- Annual Mean of SO_2 Time-series (Future vs Present)



We can see that the concentration of sulfur dioxide is gradually decreasing. The average of the historical data is 0.67 and future data is 0.34.

RESULTS

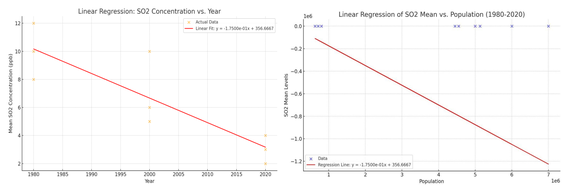
- SO_2 Per Capita by State (1980-2020)



- Population and SO_2 Levels by State (1980-2020)



- Linear Regression Models



Linear Regression: $y = mx + b$

y: the dependent variable (SO_2 Mean Levels) m: the slope of the regression line
x: the independent variable (Population) b: the y-intercept

Linear Regression for SO_2 (1980): $y = 4.4823 \times 10^{-21} x + 173.9103$
Linear Regression for SO_2 (2000): $y = 2.6044 \times 10^{-21} x + 83.0115$
Linear Regression for SO_2 (2020): $y = 4.8464 \times 10^{-22} x + 10.0103$

Correlation Coefficient > 0

CONCLUSIONS

I. Significance of Results:

- Population vs SO_2 Pollution:

Linear regression analysis reveals that population growth has no significant effect on SO_2 emissions, with correlation coefficients less than 0.01, indicating a negligible relationship. This suggests that population size alone is not a primary driver of SO_2 pollution trends.

- Population Size vs Pollution Per Person:

Interestingly, SO_2 per capita analysis indicates that more populous states exhibit lower SO_2 emissions per capita despite their larger populations. This trend likely reflects the influence of state-level environmental policies and regulations that successfully mitigate emissions, demonstrating the importance of targeted environmental governance in reducing per capita pollution.

II. Limitations:

- Technological and Industrial Changes:

Over time, technological advancements (e.g., cleaner energy technologies) and shifts in industrial activities (e.g., transitioning from coal to natural gas) have influenced SO_2 emissions. These changes occur independently of population trends and introduce complexity, making it challenging to establish direct correlations between population growth and SO_2 pollution.

- Data Availability and Temporal Factors:

Variations in available emissions data across different states and time periods may limit the scope of analysis. Additionally, temporal factors such as short-term policy changes or economic downturns could obscure long-term trends.

III. Future Work:

- Incorporating Economic, Technological, and Policy Factors:

- Objective: Future research should explore how economic growth, technological advancements, and policy interventions (e.g., the Clean Air Act, state-level emissions standards) contribute to changes in SO_2 emissions.

- Action: Develop a causal model or multivariable regression framework to analyze how these external factors influence SO_2 trends, beyond population growth alone.

- Broader Impact: Understanding the relative contributions of these factors can inform policymakers and stakeholders on effective strategies for sustainable air quality management.

- Expanding the Scope of Analysis:

Future studies could integrate other pollutants, such as NOx or $PM_{2.5}$, to provide a more comprehensive view of air quality challenges. Additionally, regional comparisons across diverse socio-economic and policy contexts could uncover best practices for emission reduction.

REFERENCES

1) Giorgi, E. (2020). Concentrations of criteria pollutants in the contiguous U.S., 1979 – 2015: Role of prediction model parsimony in integrated empirical geographic regression. *PloS One.*, *15*(2).

2) SMITH, S. J., PITCHER, H., & WIGLEY, T. M. L. (2005). Future sulfur dioxide emissions. *Climatic Change*, *73*(3), 267–318.

3) U.S. EPA. Integrated Science Assessment for Sulfur Oxides - Health Criteria (Final Report, December 2017). U.S. Environmental Protection Agency, Washington, DC, EPA/600/R-17/451, 2017.

4) U.S. EPA. Policy Assessment for the Review of the Primary NAAQS for Sulfur Oxides (Final Report, May 2018). U.S.Environmental Protection Agency, Research Triangle Park, NC, EPA-452/R-18-002, 2018.

RESEARCH POSTER PRESENTATION DESIGN © 2015
www.PosterPresentations.com